**Module I Introduction to Big Data**

Introduction, distributed file system, Big Data and its importance, Four Vs, Drivers for Big data, Big data analytics, Big data applications. Algorithms using map reduce, Matrix-Vector Multiplication by Map Reduce.

**Module II Introduction to HADOOP**

Big Data, Apache Hadoop & Hadoop Ecosystem, Moving Data in and out of Hadoop, Understanding inputs and outputs of MapReduce, Data Serialization.

**Module III HADOOP Architecture**

Hadoop Architecture, Hadoop Storage: HDFS, Common Hadoop Shell commands, Anatomy of File Write and Read, NameNode, Secondary NameNode, and DataNode, Hadoop MapReduce Paradigm, Map and Reduce tasks.

**Module IV HADOOP ecosystem**

Advanced Data Structures and Algorithm HADOOP ecosystem and yarn Hadoop ecosystem components - Schedulers - Fair and Capacity, Hadoop 2.0

**Module V New Features**

Name Node High Availability, HDFS Federation, MRv2, YARN, Running, MRv1 in YARN.

**1. Introduction to Big Data**

Big Data refers to data sets that are so large, complex, or fast-changing that traditional data processing tools can't handle them effectively. These datasets come from various sources: social media, financial transactions, IoT devices, sensors, medical imaging, etc. Big Data is important because it can reveal patterns and trends that help businesses, governments, and researchers make better decisions.
Example: Google processes over 20 petabytes of data every day for search optimization.

**2. Distributed File System (DFS)**

A Distributed File System stores data not on a single server but spread out over many machines in a network. This setup allows the system to handle huge volumes of data efficiently, provide fault tolerance (if one node fails, data is safe on another), and perform parallel processing (multiple nodes work at the same time).

Example: Hadoop Distributed File System (HDFS) splits data into blocks and stores them across nodes with replication for reliability.

**3. Importance of Big Data**

Big Data helps in understanding customer preferences, improving products, detecting fraud, predicting diseases, etc. It plays a vital role in artificial intelligence, machine learning, data-driven marketing, and scientific research. Big Data allows for real-time decision-making in critical areas like financial trading or emergency response.

**4. The Four Vs of Big Data**

These are the defining characteristics of Big Data:

• Volume – Large amounts of data measured in terabytes or petabytes.
• Velocity – Speed of data generation from various sources.
• Variety – Different data formats: text, images, video, etc.
• Veracity – Quality and trustworthiness of the data.

Some also consider a fifth V: Value – turning data into useful insights.

**5. Drivers for Big Data Growth**

Big Data has grown due to:

• Social media platforms producing massive content.
• Mobile apps and smartphones constantly generating data.
• IoT devices collecting sensor data.
• E-commerce tracking every user action.
• Business digital transformation and cloud computing.

**6. Big Data Analytics**

Big Data Analytics uses tools and techniques to clean, process, and analyze large datasets, identify patterns, correlations, and support decision-making.

Types of analytics:

• Descriptive: What happened?
• Diagnostic: Why did it happen?
• Predictive: What will happen?
• Prescriptive: What should we do?

**7. Applications of Big Data**

Big Data is applied in many sectors:

• Healthcare – disease prediction, drug development.
• Finance – fraud detection, risk assessment.

• Retail – recommendation systems, customer profiling.
• Manufacturing – predictive maintenance.
• Education – tracking student performance.
• Transportation – real-time traffic and route planning.

In today's world, there are a lot of data. Big companies utilize those data for their business growth. By analyzing this data, the useful decision can be made in various cases as discussed below:

**1. Tracking Customer Spending Habit, Shopping Behavior:** In big retails store (like Amazon, Walmart, Big Bazar etc.) management team has to keep data of customer's spending habit (in which product customer spent, in which brand they wish to spent, how frequently they spent), shopping behavior, customer's most liked product (so that they can keep those products in the store). Which product is being searched/sold most, based on that data, production/collection rate of that product get fixed.

Banking sector uses their customer's spending behavior-related data so that they can provide the offer to a particular customer to buy his particular liked product by using bank's credit or debit card with discount or cashback. By this way, they can send the right offer to the right person at the right time.

**2. Recommendation:** By tracking customer spending habit, shopping behavior, Big retails store provide a recommendation to the customer. E-commerce site like Amazon, Walmart, Flipkart does product recommendation. They track what product a customer is searching, based on that data they recommend that type of product to that customer.

As an example, suppose any customer searched bed cover on Amazon. So, Amazon got data that customer may be interested to buy bed cover. Next time when that customer will go to any google page, advertisement of various bed covers will be seen. Thus, advertisement of the right product to the right customer can be sent.

YouTube also shows recommend video based on user's previous liked, watched video type. Based on the content of a video, the user is watching, relevant advertisement is shown during video running. As an example suppose someone watching a tutorial video of Big data, then advertisement of some other big data course will be shown during that video.

**3. Smart Traffic System:** Data about the condition of the traffic of different road, collected through camera kept beside the road, at entry and exit point of the city, GPS device placed in the vehicle (Ola, Uber cab, etc.). All such data are analyzed and jam-free or less jam way, less time taking ways are recommended. Such a way smart traffic system can be built in the city by Big data analysis. One more profit is fuel consumption can be reduced.

**4. Secure Air Traffic System:** At various places of flight (like propeller etc) sensors present. These sensors capture data like the speed of flight, moisture, temperature, other

environmental condition. Based on such data analysis, an environmental parameter within flight are set up and varied.

By analyzing flight's machine-generated data, it can be estimated how long the machine can operate flawlessly when it to be replaced/repaired.

**5. Auto Driving Car:** Big data analysis helps drive a car without human interpretation. In the various spot of car camera, a sensor placed, that gather data like the size of the surrounding car, obstacle, distance from those, etc. These data are being analyzed, then various calculation like how many angles to rotate, what should be speed, when to stop, etc carried out. These calculations help to take action automatically.

**6. Virtual Personal Assistant Tool:** Big data analysis helps virtual personal assistant tool (like Siri in Apple Device, Cortana in Windows, Google Assistant in Android) to provide the answer of the various question asked by users. This tool tracks the location of the user, their local time, season, other data related to question asked, etc. Analyzing all such data, it provides an answer.

As an example, suppose one user asks "Do I need to take Umbrella?", the tool collects data like location of the user, season and weather condition at that location, then analyze these data to conclude if there is a chance of raining, then provide the answer.

**7. IoT:**

- Manufacturing company install IOT sensor into machines to collect operational data. Analyzing such data, it can be predicted how long machine will work without any problem when it requires repairing so that company can take action before the situation when machine facing a lot of issues or gets totally down. Thus, the cost to replace the whole machine can be saved.
- In the Healthcare field, Big data is providing a significant contribution. Using big data tool, data regarding patient experience is collected and is used by doctors to give better treatment. IoT device can sense a symptom of probable coming disease in the human body and prevent it from giving advance treatment. IoT Sensor placed near-patient, new-born baby constantly keeps track of various health condition like heart bit rate, blood presser, etc. Whenever any parameter crosses the safe limit, an alarm sent to a doctor, so that they can take step remotely very soon.

**8. Education Sector:** Online educational course conducting organization utilize big data to search candidate, interested in that course. If someone searches for YouTube tutorial video on a subject, then online or offline course provider organization on that subject send ad online to that person about their course.

**9. Energy Sector:** Smart electric meter read consumed power every 15 minutes and sends this read data to the server, where data analyzed and it can be estimated what is the time in a day when the power load is less throughout the city. By this system manufacturing unit or housekeeper are suggested the time when they should drive their heavy machine in the night time when power load less to enjoy less electricity bill.

**10. Media and Entertainment Sector:** Media and entertainment service providing company like Netflix, Amazon Prime, Spotify do analysis on data collected from their

users. Data like what type of video, music users are watching, listening most, how long users are spending on site, etc are collected and analyzed to set the next business strategy.

**8. Algorithms Using MapReduce**

MapReduce is a programming model introduced by Google for processing large-scale data in parallel:

• Map Function – breaks down input into key-value pairs.
• Shuffle & Sort – groups intermediate data by key.
• Reduce Function – processes grouped data for final output.

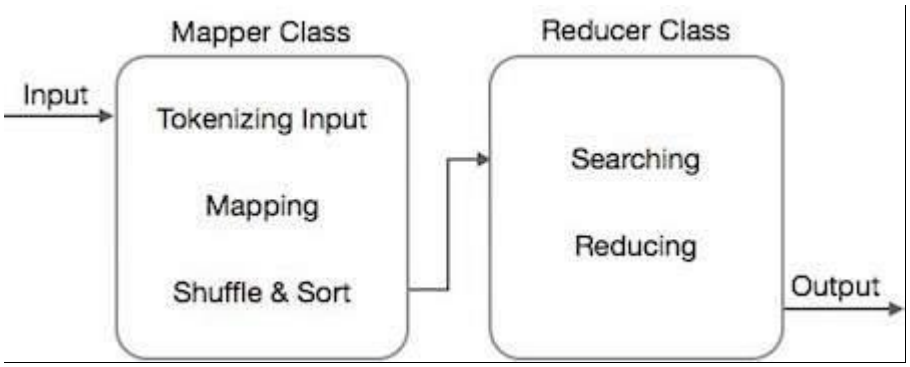
Example: Used in counting word frequency, sorting, filtering data.

The MapReduce algorithm contains two important tasks, namely Map and Reduce.

The map task is done by means of Mapper Class

The reduce task is done by means of Reducer Class.

Mapper class takes the input, tokenizes it, maps and sorts it. The output of Mapper class is used as input by Reducer class, which in turn searches matching pairs and reduces them.




**Mapper Reducer Class**

MapReduce implements various mathematical algorithms to divide a task into small parts and assign them to multiple systems. In technical terms, MapReduce algorithm helps in sending the Map & Reduce tasks to appropriate servers in a cluster.

These mathematical algorithms may include the following −

**Sorting**

**Searching**

**Indexing**

**TF-IDF**

**Sorting**

**Sorting** is one of the basic MapReduce algorithms to process and analyze data. MapReduce implements sorting algorithm to automatically sort the output key-value pairs from the mapper by their keys.

Sorting methods are implemented in the mapper class itself.

In the Shuffle and Sort phase, after tokenizing the values in the mapper class, the **Context** class (user-defined class) collects the matching valued keys as a collection.

To collect similar key-value pairs (intermediate keys), the Mapper class takes the help of **RawComparator** class to sort the key-value pairs.

The set of intermediate key-value pairs for a given Reducer is automatically sorted by Hadoop to form key-values (K2, {V2, V2, }) before they are presented to the Reducer

## Searching

Searching plays an important role in MapReduce algorithm. It helps in the combiner phase (optional) and in the Reducer phase.we can see how Searching works with the help of an example.

## Example

The following example shows how MapReduce employs Searching algorithm to find out the details of the employee who draws the highest salary in a given employee dataset.

Let us assume we have employee data in four different files − A, B, C, and D. Let us also assume there are duplicate employee records in all four files because of importing the employee data from all database tables repeatedly. See the following illustration.

| name, salary |
|---|
| satish, 26000 |
| Krishna, 25000 |
| Satishk, 15000 |
| Raju, 10000 |

| name, salary |
|---|
| gopal, 50000 |
| Krishna, 25000 |
| Satishk, 15000 |
| Raju, 10000 |

| name, salary |
|---|
| satish, 26000 |
| kiran, 45000 |
| Satishk, 15000 |
| Raju, 10000 |

| name, salary |
|---|
| satish, 26000 |
| Krishna, 25000 |
| manisha, 45000 |
| Raju, 10000 |

**The Map phase** processes each input file and provides the employee data in key-value pairs (<k, v> : <emp name, salary>). See the following illustration.

**The combiner phase**

.(searching technique) will accept the input from the Map phase as a key-value pair with employee name and salary. Using searching technique, the combiner will check all the employee salary to find the highest salaried employee in each file. See the following snippet.

```
<k: employee name, v: salary>
Max= the salary of an first employee. Treated as max salary

if(v(second employee).salary > Max){
   Max = v(salary);
}

else{
   Continue checking;
}
```

The expected result is as follows −

| | | | |
|---|---|---|---|
| <satish, 26000> | <gopal, 50000> | <kiran, 45000> | <manisha, 45000> |

**Reducer phase** − Form each file, you will find the highest salaried employee. To avoid redundancy, check all the <k, v> pairs and eliminate duplicate entries, if any. The same algorithm is used in between the four <k, v> pairs, which are coming from four input files. The final output should be as follows −

<gopal, 50000>

## Indexing

Normally indexing is used to point to a particular data and its address. It performs batch indexing on the input files for a particular Mapper.

The indexing technique that is normally used in MapReduce is known as **inverted index.** Search engines like Google and Bing use inverted indexing technique.

Here we can see  how Indexing works with the help of a simple example.

## Example

The following text is the input for inverted indexing. Here T[0], T[1], and t[2] are the file names and their content are in double quotes.

T[0] = "it is what it is"

T[1] = "what is it"

T[2] = "it is a banana"

After applying the Indexing algorithm, we get the following output −

"a": {2}

"banana": {2}

"is": {0, 1, 2}

"it": {0, 1, 2}

"what": {0, 1}

Here "a": {2} implies the term "a" appears in the T[2] file. Similarly, "is": {0, 1, 2} implies the term "is" appears in the files T[0], T[1], and T[2].

## TF-IDF

TF-IDF is a text processing algorithm which is short for Term Frequency − Inverse Document Frequency. It is one of the common web analysis algorithms. Here, the term 'frequency' refers to the number of times a term appears in a document.

## Term Frequency (TF)

It measures how frequently a particular term occurs in a document. It is calculated by the number of times a word appears in a document divided by the total number of words in that document.

TF(the) = (Number of times term the the appears in a document) / (Total number of terms in the document)

## Inverse Document Frequency (IDF)

It measures the importance of a term. It is calculated by the number of documents in the text database divided by the number of documents where a specific term appears.

While computing TF, all the terms are considered equally important. That means, TF counts the term frequency for normal words like is, a, what, etc. Thus we need to know the frequent terms while scaling up the rare ones, by computing the following −

IDF(the) = log_e(Total number of documents / Number of documents with term the in it).

The algorithm is explained below with the help of a small example.

## Example

Consider a document containing 1000 words, wherein the word **hive** appears 50 times. The TF for **hive** is then (50 / 1000) = 0.05.

Now, assume we have 10 million documents and the word **hive** appears in 1000 of these. Then, the IDF is calculated as log(10,000,000 / 1,000) = 4.

The TF-IDF weight is the product of these quantities − 0.05 4 = 0.20.

### 9. Matrix-Vector Multiplication Using MapReduce

Goal: Compute b = Ax where A is a matrix and x is a vector.
• Map Phase – Each row of matrix A is multiplied with vector x.
• Emit key-value pairs (row index, partial product).
• Reduce Phase – Sum partial products to get final output for each row.
Used in machine learning and distributed computing.

### Matrix Multiplication With 1 MapReduce Step

MapReduce is a technique in which a huge program is subdivided into small tasks and run parallelly to make computation faster, save time, and mostly used in distributed systems. It has 2 important parts:

- **Mapper:** It takes raw data input and organizes into key, value pairs. For example, In a dictionary, you search for the word "Data" and its associated meaning is "facts and statistics collected together for reference or analysis". Here the Key is *Data* and the **Value** associated with is *facts and statistics collected together for reference or analysis.*
- **Reducer:** It is responsible for processing data in parallel and produce final output.

Let us consider the matrix multiplication example to visualize MapReduce. Consider the following matrix:

2x2 matrices A and B

Here matrix A is a 2x2 matrix which means the number of rows(i)=2 and the number of

columns(j)=2. Matrix B is also a 2x2 matrix where number of rows(j)=2 and number of columns(k)=2. Each cell of the matrix is labelled as Aij and Bij. Ex. element 3 in matrix A is called A21 i.e. 2nd-row 1st column. Now One step matrix multiplication has 1 mapper and 1 reducer. The Formula is:

*Mapper for Matrix A (k, v)=((i, k), (A, j, Aij)) for all k*
*Mapper for Matrix B (k, v)=((i, k), (B, j, Bjk)) for all i*

Therefore computing the mapper for Matrix A:

```
# k, i, j computes the number of times it occurs.
# Here all are 2, therefore when k=1, i can have
# 2 values 1 & 2, each case can have 2 further
# values of j=1 and j=2. Substituting all values
# in formula

k=1  i=1  j=1   ((1, 1), (A, 1, 1))
        j=2   ((1, 1), (A, 2, 2))
    i=2  j=1   ((2, 1), (A, 1, 3))
        j=2   ((2, 1), (A, 2, 4))

k=2  i=1  j=1   ((1, 2), (A, 1, 1))
        j=2   ((1, 2), (A, 2, 2))
    i=2  j=1   ((2, 2), (A, 1, 3))
        j=2   ((2, 2), (A, 2, 4))
```

Computing the mapper for Matrix B

```
i=1  j=1  k=1   ((1, 1), (B, 1, 5))
        k=2   ((1, 2), (B, 1, 6))
    j=2  k=1   ((1, 1), (B, 2, 7))
        k=2   ((1, 2), (B, 2, 8))

i=2  j=1  k=1   ((2, 1), (B, 1, 5))
        k=2   ((2, 2), (B, 1, 6))
    j=2  k=1   ((2, 1), (B, 2, 7))
        k=2   ((2, 2), (B, 2, 8))
```

**The formula for Reducer is:**

*Reducer(k, v)=(i, k)=>Make sorted Alist and Blist*
*(i, k) => Summation (Aij * Bjk)) for j*
*Output =>((i, k), sum)*

Therefore computing the reducer:

```
# We can observe from Mapper computation
# that 4 pairs are common (1, 1), (1, 2),
# (2, 1) and (2, 2)
# Make a list separate for Matrix A &
# B with adjoining values taken from
# Mapper step above:

(1, 1) =>Alist ={(A, 1, 1), (A, 2, 2)}
        Blist ={(B, 1, 5), (B, 2, 7)}
        Now Aij x Bjk: [(1*5) + (2*7)] =19 -------(i)

(1, 2) =>Alist ={(A, 1, 1), (A, 2, 2)}
        Blist ={(B, 1, 6), (B, 2, 8)}
        Now Aij x Bjk: [(1*6) + (2*8)] =22 -------(ii)

(2, 1) =>Alist ={(A, 1, 3), (A, 2, 4)}
        Blist ={(B, 1, 5), (B, 2, 7)}
        Now Aij x Bjk: [(3*5) + (4*7)] =43 -------(iii)

(2, 2) =>Alist ={(A, 1, 3), (A, 2, 4)}
        Blist ={(B, 1, 6), (B, 2, 8)}
        Now Aij x Bjk: [(3*6) + (4*8)] =50 -------(iv)

From (i), (ii), (iii) and (iv) we conclude that
((1, 1), 19)
((1, 2), 22)
```

((2, 1), 43)

((2, 2), 50)

Therefore the Final Matrix is:

$$\begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$ Final output of Matrix multiplication.

Comment
More info